195833: sandstone, Sydney Heads Pass

(Yelma Formation, Tooloo Group, Earaheedy Basin, Capricorn Orogen)

Location and sampling

Stanley (SG 51-6), Earaheedy (3246)
MGA Zone 51, 377456E 7178522N

Sampled on 24 July 2011

This sample was collected from a boulder adjacent to an outcrop (Fig. 1) on the east side of Sydney Head Road, where it passes through the Mudan Hills at Sydney Heads Pass. The sample site is located about 22 km northeast of Earaheedy Homestead, 7.9 km southeast of Imbin Rockhole, and 3.2 km east of Mount Evelyn.

Geological context

The unit sampled is a quartz sandstone of the Yelma Formation, the basal unit of the Tooloo Group of the Earaheedy Basin (Pirajno et al., 2009). The formation consists of sandstone, shale and carbonate rocks, and minor siltstone and conglomerate (Hocking et al., 2020). The 1990–1949 Ma Yelma Formation unconformably overlies Archean basement rocks of the Yilgarn Craton and is overlain conformably by the Sweetwaters Well Dolomite or disconformably by the Frere Formation (Pirajno et al., 2009, Hocking et al., 2020). A sample of the unconformably overlying, late Paleoproterozoic to early Mesoproterozoic Sydney Heads Pass Conglomerate (Hocking, 2013) about 0.5 km to the northeast has a maximum depositional age of 2040 ± 17 Ma (GSWA 205570, Lu et al., 2018).

Petrographic description

The sample is a fine-grained quartz sandstone, consisting of about 98% quartz, and trace amounts of chert fragments, feldspar, and zircon. Quartz grains are 0.3 to 0.7 mm in diameter. The interlocking, irregular shapes of the grains and the lack of primary sedimentary features suggest that the rock is recrystallized. A detailed petrographic description is not available for this sample.

Zircon morphology

Images or a description of the zircons for this sample are not available.

Analytical details

This sample was analysed on 22–23 August, 9 September, and 15–16 November 2012, using SHRIMP-B. Analyses 1.1 to 59.1 (spot numbers 1–64) were obtained during the first session, together with 17 analyses of the BR266 standard. Significant drift of standard $^{238}U/^{206}Pb^*$ dates during the session was addressed by fitting a LOWESS curve (Cleveland, 1979) with a smoothing window of five analyses, implemented using the program Squid 2.50 (Ludwig, 2009; Wingate et al., 2022). The analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.83% (1σ), which is included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Analyses 60.1 to 73.1 (spot numbers 65–78) were obtained during the second session, together with 10 analyses of the BR266 standard, of which nine analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.99% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.36% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the second session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.33%. Analyses 74.1 to 53.3 (spot numbers 79–124) were obtained during the third session, together with 12 analyses of the BR266 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.83% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

A total of 124 analyses were obtained from 118 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 2), and a probability density diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 2). Twenty-six analyses are >5% discordant. The dates obtained from these 26 analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 98 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 1908 ± 14 Ma (1σ).

Group S comprises 96 analyses of 94 zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3008–1980 Ma.

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 1850 ± 17 Ma (1σ).

Figure 1. Outcrop image for sample 195833: sandstone, Sydney Heads Pass

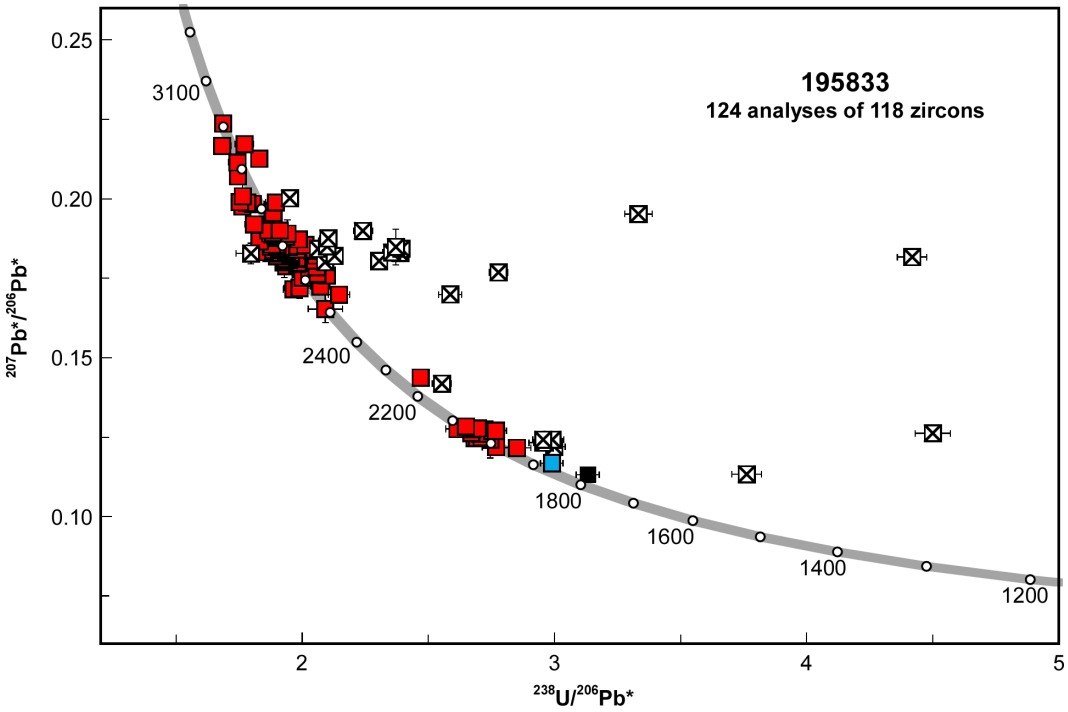

Figure 2. U–Pb analytical data for sample 195833: sandstone, Sydney Heads Pass. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); black square indicates Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%). One highly discordant analysis in Group D is not shown

Table 1. Ion microprobe analytical results for zircons from sample 195833: sandstone, Sydney Heads Pass

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
Y	59	53.2	88	86	1.01	0.042	2.988	0.045	0.11715	0.00088	2.990	0.045	0.11678	0.00091	1860	25	1908	14	2.5
S	17	14.2	34	26	0.81	-0.053	2.853	0.056	0.12118	0.00143	2.852	0.056	0.12164	0.00151	1938	33	1980	22	2.2
S	7	1.2	32	64	2.04	0.110	2.766	0.055	0.12278	0.00147	2.769	0.055	0.12181	0.00162	1987	34	1983	24	-0.2
S	65	60.1	84	73	0.89	0.198	2.741	0.048	0.12592	0.00577	2.746	0.048	0.12418	0.00581	2002	31	2017	83	0.5
S	96	91.1	111	38	0.36	0.034	2.682	0.042	0.12500	0.00082	2.683	0.042	0.12470	0.00084	2042	27	2025	12	-0.9
S	105	100.1	119	139	1.21	0.079	2.691	0.041	0.12544	0.00080	2.693	0.041	0.12475	0.00086	2036	27	2025	12	-0.5
S	46	43.1	78	40	0.53	0.085	2.709	0.042	0.12560	0.00093	2.711	0.042	0.12486	0.00100	2024	27	2027	14	0.1
S	43	40.1	132	84	0.66	0.048	2.703	0.038	0.12649	0.00071	2.705	0.038	0.12606	0.00074	2028	25	2044	10	0.8
S	8	7.1	144	181	1.30	-0.012	2.671	0.038	0.12618	0.00072	2.671	0.038	0.12629	0.00072	2050	25	2047	10	-0.2
S	1	1.1	31	62	2.08	-0.172	2.759	0.056	0.12545	0.00185	2.754	0.056	0.12697	0.00204	1997	36	2056	28	2.9
S	52	49.1	105	108	1.06	0.000	2.768	0.041	0.12706	0.00083	2.768	0.041	0.12706	0.00083	1988	25	2058	12	3.4
S	88	83.1	95	77	0.83	0.120	2.671	0.043	0.12855	0.00091	2.674	0.043	0.12750	0.00101	2048	29	2064	14	0.8
S	56	51.2	45	37	0.84	-0.031	2.617	0.046	0.12732	0.00113	2.616	0.046	0.12759	0.00117	2087	32	2065	16	-1.0
S	45	42.1	45	57	1.33	-0.136	2.675	0.048	0.12650	0.00120	2.671	0.048	0.12769	0.00134	2050	32	2066	19	0.8
S	54	51.1	60	60	1.04	0.103	2.693	0.044	0.12862	0.00105	2.696	0.044	0.12771	0.00114	2034	29	2067	16	1.6
S	122	117.1	136	129	0.98	0.136	2.645	0.040	0.12952	0.00076	2.649	0.040	0.12832	0.00085	2065	27	2075	12	0.5
S	40	37.1	174	48	0.28	-0.040	2.471	0.034	0.14337	0.00076	2.470	0.034	0.14372	0.00077	2192	26	2272	9	3.6
S	12	11.1	10	10	1.09	0.910	2.073	0.067	0.17345	0.00297	2.092	0.068	0.16533	0.00431	2519	70	2511	44	-0.3
S	25	23.1	35	49	1.47	0.173	2.142	0.042	0.17134	0.00158	2.146	0.042	0.16980	0.00173	2466	41	2556	17	3.5
S	2	2.1	27	31	1.16	0.045	1.965	0.041	0.17188	0.00173	1.966	0.041	0.17148	0.00178	2651	47	2572	17	-3.1
S	117	112.1	12	7	0.62	0.415	1.982	0.062	0.17562	0.00264	1.990	0.063	0.17192	0.00324	2624	70	2576	31	-1.9
S	61	56.1	45	41	0.95	0.052	2.071	0.037	0.17287	0.00134	2.072	0.037	0.17241	0.00138	2539	38	2581	13	1.6
S	86	81.1	38	43	1.19	-0.068	2.002	0.041	0.17453	0.00154	2.001	0.041	0.17514	0.00160	2613	45	2607	15	-0.2
S	33	30.1	160	70	0.45	0.000	2.097	0.029	0.17583	0.00075	2.097	0.029	0.17583	0.00075	2514	29	2614	7	3.8
S	72	67.1	11	14	1.30	-0.199	2.032	0.064	0.17430	0.00260	2.028	0.064	0.17608	0.00288	2584	68	2616	27	1.0
S	16	15.1	18	16	0.89	0.187	2.057	0.051	0.17799	0.00209	2.061	0.051	0.17632	0.00231	2550	53	2619	22	2.6
S	31	28.1	75	73	1.00	0.063	2.056	0.032	0.17905	0.00106	2.057	0.032	0.17849	0.00109	2554	34	2639	10	3.2
S	114	109.1	86	99	1.20	0.134	1.935	0.032	0.18003	0.00103	1.937	0.032	0.17884	0.00111	2683	37	2642	10	-1.6
S	81	76.1	34	43	1.32	0.373	1.974	0.042	0.18281	0.00163	1.981	0.042	0.17948	0.00195	2634	46	2648	18	0.5
S	67	62.1	29	36	1.28	0.253	1.982	0.045	0.18208	0.00176	1.987	0.046	0.17982	0.00199	2628	50	2651	18	0.7
S	109	104.1	82	97	1.21	0.256	1.922	0.032	0.18232	0.00110	1.927	0.032	0.18003	0.00125	2695	38	2653	12	-1.6

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	ƒ204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	123	118.1	56	55	1.02	0.286	1.925	0.035	0.18278	0.00496	1.930	0.035	0.18023	0.00503	2691	41	2655	46	-1.3
S	27	25.1	153	111	0.75	0.062	2.045	0.028	0.18109	0.00075	2.046	0.028	0.18054	0.00078	2565	30	2658	7	3.5
S	83	78.1	72	43	0.62	0.247	1.941	0.034	0.18283	0.00117	1.946	0.034	0.18063	0.00132	2673	39	2659	12	-0.6
S	98	93.1	55	32	0.60	0.200	1.932	0.036	0.18261	0.00134	1.936	0.036	0.18083	0.00149	2684	42	2661	14	-0.9
S	39	36.1	89	57	0.66	0.000	1.951	0.030	0.18095	0.00093	1.951	0.030	0.18095	0.00093	2668	34	2662	9	-0.2
S	119	114.1	173	124	0.74	0.048	1.986	0.029	0.18145	0.00348	1.987	0.029	0.18101	0.00349	2627	32	2662	32	1.3
S	66	61.1	53	33	0.64	0.242	1.931	0.037	0.18334	0.00145	1.936	0.037	0.18117	0.00158	2685	42	2664	14	-1.0
S	99	94.1	111	117	1.09	0.154	1.977	0.031	0.18283	0.00093	1.980	0.031	0.18145	0.00101	2636	35	2666	9	1.1
S	70	65.1	52	37	0.73	0.282	1.931	0.037	0.18433	0.00125	1.937	0.037	0.18181	0.00143	2683	43	2669	13	-0.7
S	87	82.1	143	131	0.94	0.079	1.955	0.029	0.18251	0.00080	1.957	0.029	0.18181	0.00083	2661	33	2669	8	0.3
S	3	3.1	118	104	0.92	0.000	1.901	0.028	0.18191	0.00087	1.901	0.028	0.18191	0.00087	2724	33	2670	8	-2.0
S	26	24.1	75	70	0.96	0.082	2.054	0.032	0.18295	0.00102	2.056	0.032	0.18222	0.00107	2555	33	2673	10	4.4
S	107	102.1	51	42	0.83	0.022	1.953	0.036	0.18266	0.00126	1.953	0.036	0.18246	0.00128	2665	41	2675	12	0.4
S	103	98.1	90	104	1.20	0.099	1.928	0.032	0.18357	0.00102	1.930	0.032	0.18269	0.00107	2692	37	2677	10	-0.5
S	55	52.1	11	11	1.06	-0.098	1.910	0.057	0.18193	0.00269	1.908	0.057	0.18281	0.00283	2717	68	2678	26	-1.4
S	53	50.1	83	53	0.67	0.000	1.941	0.030	0.18286	0.00096	1.941	0.030	0.18286	0.00096	2679	34	2679	9	0.0
S	110	105.1	103	113	1.13	0.078	1.928	0.031	0.18360	0.00097	1.930	0.031	0.18291	0.00101	2692	36	2679	9	-0.5
S	92	87.1	44	25	0.58	0.224	1.951	0.038	0.18493	0.00143	1.955	0.038	0.18293	0.00160	2663	43	2680	14	0.6
S	104	99.1	31	19	0.64	0.125	1.876	0.042	0.18411	0.00174	1.878	0.042	0.18299	0.00185	2752	51	2680	17	-2.7
S	111	106.1	60	21	0.37	0.073	1.919	0.034	0.18365	0.00115	1.921	0.034	0.18300	0.00119	2702	40	2680	11	-0.8
S	58	54.1	61	60	1.02	0.000	1.889	0.032	0.18302	0.00122	1.889	0.032	0.18302	0.00122	2738	38	2680	11	-2.2
S	64	59.1	14	7	0.51	0.078	1.850	0.050	0.18390	0.00240	1.852	0.050	0.18320	0.00250	2784	63	2682	23	-3.8
S	74	69.1	39	51	1.37	0.138	1.936	0.040	0.18454	0.00145	1.939	0.040	0.18331	0.00155	2682	46	2683	14	-0.1
S	38	35.1	109	65	0.62	-0.055	1.939	0.028	0.18308	0.00082	1.938	0.028	0.18357	0.00085	2682	32	2685	8	0.1
S	95	90.1	47	42	0.93	0.428	2.033	0.044	0.18749	0.00180	2.042	0.044	0.18367	0.00218	2569	46	2686	20	4.4
S	91	86.1	36	26	0.75	0.109	1.883	0.040	0.18474	0.00163	1.885	0.040	0.18377	0.00172	2743	48	2687	15	-2.1
S	13	12.1	94	58	0.64	0.000	2.008	0.030	0.18393	0.00094	2.008	0.030	0.18393	0.00094	2605	33	2689	8	3.1
S	113	108.1	98	67	0.71	0.085	1.944	0.031	0.18473	0.00095	1.945	0.031	0.18396	0.00100	2674	36	2689	9	0.6
S	11	10.1	144	138	0.99	0.015	1.984	0.028	0.18414	0.00095	1.984	0.028	0.18400	0.00095	2631	31	2689	9	2.2
S	80	75.1	91	55	0.62	0.096	1.977	0.032	0.18489	0.00100	1.979	0.032	0.18403	0.00106	2637	36	2690	9	2.0
S	106	101.1	38	29	0.78	0.029	1.914	0.038	0.18478	0.00155	1.914	0.039	0.18452	0.00157	2709	45	2694	14	-0.6
S	78	73.1	109	83	0.79	0.145	1.905	0.032	0.18586	0.00094	1.908	0.032	0.18457	0.00100	2716	38	2694	9	-1.0

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$f204$ (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	42	39.1	146	182	1.29	-0.007	1.947	0.027	0.18474	0.00073	1.947	0.027	0.18480	0.00073	2672	31	2696	7	0.9
S	73	68.1	65	62	0.98	0.213	1.893	0.116	0.18672	0.00111	1.897	0.116	0.18481	0.00122	2729	144	2697	11	-1.4
S	68	63.1	35	17	0.50	-0.062	1.878	0.040	0.18441	0.00153	1.876	0.040	0.18496	0.00158	2754	49	2698	14	-2.3
S	49	46.1	82	102	1.28	-0.014	1.980	0.031	0.18511	0.00103	1.980	0.031	0.18523	0.00103	2636	34	2700	9	2.4
S	48	45.1	86	64	0.76	0.012	2.011	0.031	0.18562	0.00097	2.012	0.031	0.18551	0.00097	2601	33	2703	9	3.8
S	79	74.1	32	12	0.38	0.163	1.898	0.042	0.18702	0.00174	1.901	0.042	0.18556	0.00189	2725	51	2703	17	-0.8
S	102	97.1	77	85	1.14	0.049	1.879	0.032	0.18607	0.00111	1.880	0.032	0.18564	0.00114	2749	39	2704	10	-1.7
S	5	5.1	93	42	0.46	0.000	1.831	0.028	0.18651	0.00095	1.831	0.028	0.18651	0.00095	2809	35	2712	8	-3.6
S	97	92.1	84	88	1.08	0.113	1.984	0.033	0.18822	0.00111	1.986	0.033	0.18721	0.00117	2629	37	2718	10	3.3
S	118	113.1	74	37	0.52	0.089	1.896	0.032	0.18830	0.00106	1.898	0.032	0.18751	0.00111	2729	38	2720	10	-0.3
S	75	70.1	60	52	0.90	0.190	1.903	0.036	0.18956	0.00120	1.906	0.036	0.18786	0.00130	2719	42	2724	11	0.0
S	35	32.1	61	35	0.59	0.135	1.935	0.032	0.18932	0.00115	1.937	0.032	0.18812	0.00123	2683	36	2726	11	1.6
S	41	38.1	85	28	0.34	0.089	1.891	0.029	0.18915	0.00100	1.893	0.029	0.18836	0.00105	2734	35	2728	9	-0.2
S	121	116.1	58	29	0.52	0.099	1.881	0.034	0.18963	0.00123	1.883	0.034	0.18874	0.00129	2746	41	2731	11	-0.5
S	82	77.1	94	52	0.56	0.080	1.939	0.031	0.18969	0.00439	1.941	0.031	0.18897	0.00441	2679	36	2733	38	2.0
S	69	64.1	71	55	0.81	0.029	1.873	0.034	0.18929	0.00111	1.873	0.034	0.18903	0.00112	2758	41	2734	10	-1.1
S	101	96.1	71	51	0.74	0.000	1.884	0.033	0.18921	0.00118	1.884	0.033	0.18921	0.00118	2745	40	2735	10	-0.3
S	34	31.1	19	14	0.75	0.000	1.865	0.046	0.19000	0.00213	1.865	0.046	0.19000	0.00213	2767	57	2742	18	-0.9
S	71	66.1	22	10	0.49	-0.215	1.915	0.046	0.18807	0.00185	1.911	0.046	0.19000	0.00203	2714	54	2742	18	0.8
S	77	72.1	43	40	0.97	-0.071	1.811	0.036	0.19131	0.00141	1.810	0.036	0.19195	0.00145	2836	47	2759	12	-3.0
S	36	33.1	83	106	1.32	-0.090	1.884	0.029	0.19485	0.00103	1.883	0.029	0.19565	0.00107	2746	35	2790	9	1.6
S	63	58.1	46	26	0.59	0.000	1.886	0.033	0.19566	0.00128	1.886	0.033	0.19566	0.00128	2742	40	2790	11	1.7
S	6	6.1	35	29	0.84	0.057	1.761	0.034	0.19809	0.00154	1.762	0.034	0.19758	0.00158	2898	46	2806	13	-3.3
S	100	95.1	19	12	0.63	0.222	1.799	0.047	0.20032	0.00214	1.803	0.047	0.19835	0.00236	2844	62	2813	19	-1.1
S	112	107.1	51	22	0.44	-0.045	1.783	0.039	0.19837	0.00135	1.782	0.039	0.19876	0.00138	2871	52	2816	11	-2.0
S	94	89.1	47	50	1.10	0.172	1.891	0.039	0.20041	0.00170	1.895	0.039	0.19887	0.00184	2732	47	2817	15	3.0
S	84	79.1	95	31	0.34	0.152	1.750	0.029	0.20025	0.00108	1.753	0.029	0.19890	0.00115	2910	39	2817	9	-3.3
S	90	85.1	72	33	0.47	0.135	1.762	0.031	0.20198	0.00478	1.764	0.031	0.20078	0.00481	2895	41	2833	39	-2.2
S	10	9.1	35	9	0.27	0.085	1.743	0.034	0.20781	0.00162	1.745	0.034	0.20705	0.00168	2921	47	2883	13	-1.3
S	116	111.1	42	27	0.66	0.164	1.740	0.034	0.21290	0.00172	1.743	0.034	0.21144	0.00181	2923	46	2917	14	-0.2
S	93	88.1	68	30	0.46	0.042	1.829	0.034	0.21300	0.00140	1.830	0.034	0.21262	0.00143	2811	43	2926	11	3.9
S	76	71.1	36	28	0.80	0.024	1.683	0.035	0.21685	0.00153	1.683	0.035	0.21664	0.00154	3006	51	2956	11	-1.9

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	51	48.1	27	11	0.43	-0.030	1.772	0.036	0.21693	0.00173	1.771	0.036	0.21720	0.00175	2886	49	2960	13	2.5
S	115	110.1	79	56	0.73	0.025	1.686	0.028	0.22393	0.00112	1.686	0.028	0.22371	0.00113	3002	41	3008	8	0.2
P	57	53.1	106	98	0.95	0.407	3.120	0.046	0.11668	0.00077	3.133	0.046	0.11312	0.00106	1786	23	1850	17	3.5
D	124	53.3	114	131	1.19	0.333	3.750	0.058	0.11623	0.00100	3.763	0.059	0.11332	0.00131	1519	21	1853	21	18.0
D	28	26.1	120	65	0.56	-0.042	3.002	0.043	0.12155	0.00074	3.000	0.043	0.12192	0.00077	1854	23	1984	11	6.6
D	15	14.1	33	28	0.88	-0.051	2.958	0.058	0.12279	0.00142	2.957	0.058	0.12323	0.00148	1878	33	2004	21	6.3
D	29	26.2	117	63	0.56	-0.015	2.961	0.043	0.12311	0.00078	2.961	0.043	0.12324	0.00080	1876	24	2004	11	6.4
D	89	84.1	145	126	0.90	0.060	2.990	0.045	0.12473	0.00077	2.992	0.045	0.12421	0.00082	1859	24	2017	12	7.9
D	30	27.1	163	155	0.98	0.023	2.956	0.041	0.12443	0.00069	2.956	0.041	0.12422	0.00070	1878	23	2018	10	6.9
D	21	19.1	155	158	1.05	0.192	4.492	0.069	0.12792	0.00079	4.501	0.069	0.12623	0.00092	1293	18	2046	13	36.8
D	14	13.1	290	331	1.18	0.177	6.931	0.256	0.12972	0.00071	6.944	0.256	0.12816	0.00082	867	31	2073	11	58.2
D	50	47.1	88	216	2.53	-0.013	2.554	0.038	0.14164	0.00080	2.554	0.038	0.14176	0.00081	2130	27	2249	10	5.3
D	60	55.1	50	58	1.18	0.074	2.586	0.046	0.17056	0.00157	2.587	0.046	0.16990	0.00164	2107	33	2557	16	17.6
D	32	29.1	200	157	0.81	0.073	2.777	0.037	0.17750	0.00068	2.779	0.037	0.17684	0.00071	1982	23	2623	7	24.5
D	19	17.1	65	125	2.00	-0.036	2.092	0.034	0.17977	0.00114	2.091	0.034	0.18009	0.00117	2519	35	2654	11	5.1
D	85	80.1	191	225	1.22	0.068	2.304	0.033	0.18101	0.00076	2.305	0.033	0.18041	0.00079	2323	28	2657	7	12.6
D	9	8.1	257	175	0.70	0.269	4.407	0.057	0.18409	0.00076	4.419	0.058	0.18169	0.00093	1315	16	2668	9	50.7
D	37	34.1	96	119	1.28	0.012	2.129	0.032	0.18215	0.00094	2.129	0.032	0.18205	0.00095	2482	31	2672	9	7.1
D	108	103.1	13	14	1.12	0.392	1.789	0.058	0.18627	0.00269	1.797	0.059	0.18277	0.00322	2853	77	2678	29	-6.5
D	4	4.1	189	58	0.32	0.007	2.389	0.032	0.18301	0.00072	2.389	0.032	0.18295	0.00072	2254	26	2680	7	15.9
D	47	44.1	103	125	1.26	0.026	2.355	0.035	0.18334	0.00100	2.356	0.035	0.18311	0.00102	2281	29	2681	9	14.9
D	62	57.1	113	115	1.05	-0.011	2.393	0.035	0.18418	0.00091	2.393	0.035	0.18427	0.00091	2251	28	2692	8	16.4
D	44	41.1	67	53	0.82	-0.014	2.063	0.033	0.18416	0.00105	2.063	0.033	0.18429	0.00105	2548	34	2692	9	5.3
D	120	115.1	49	47	1.00	0.171	2.368	0.045	0.18635	0.00553	2.372	0.045	0.18483	0.00558	2268	37	2697	50	15.9
D	18	16.1	66	44	0.69	-0.019	2.100	0.035	0.18488	0.00120	2.099	0.035	0.18505	0.00121	2511	35	2699	11	6.9
D	23	21.1	53	36	0.70	0.242	2.098	0.035	0.18970	0.00117	2.103	0.035	0.18754	0.00131	2508	35	2721	11	7.8
D	24	22.1	52	45	0.90	-0.045	2.242	0.038	0.18948	0.00160	2.241	0.038	0.18988	0.00163	2379	34	2741	14	13.2
D	22	20.1	59	97	1.68	0.537	3.316	0.054	0.19994	0.00161	3.334	0.055	0.19516	0.00197	1691	25	2786	17	39.3
D	20	18.1	137	109	0.82	0.041	1.950	0.027	0.20048	0.00078	1.951	0.027	0.20011	0.00079	2668	31	2827	6	5.6

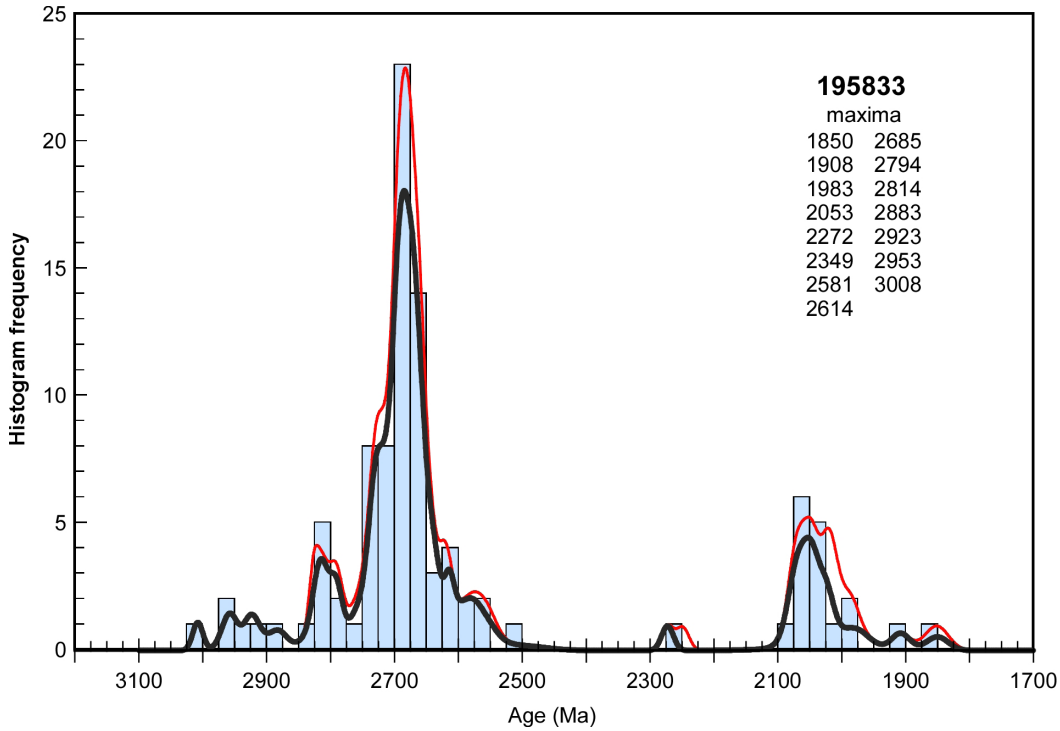

Figure 3. Probability density diagram and histogram for sample 195833: sandstone, Sydney Heads Pass. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) only include data with discordance <5% (98 analyses of 94 zircons). Thin curve includes all data (124 analyses of 118 zircons)

It is possible that all of the analyses in Groups Y and S are of unmodified detrital zircons, in which case the date of 1908 ± 14 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sandstone. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2035 ± 15 Ma (MSWD = 2.1) for the youngest 10 analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 2814, 2794, 2685, 2614, 2581, 2053, and 1983 Ma, based on contributions from approximately 6, 8, 34, 8, 9, 11, and 3 analyses, respectively (Fig. 3). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

The date of 1850 ± 17 Ma (1σ) for the single analysis in Group P is interpreted to reflect minor loss of radiogenic Pb. A second analysis of this zircon yielded an older date of 1908 ± 14 Ma (1σ).

References

Cleveland, WS 1979, Robust, locally weighted regression and smoothing scatter plots: Journal of the American Statistical Association, v. 74, p. 829–836.

Hocking, RM 2013, Sydney Heads Pass Conglomerate (P_-_sh-sc): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 10 December 2021, <www.dmp.wa.gov.au/ens>.

Hocking, RM, Jones, JA and Pirajno, F 2020, Yelma Formation (P_-ETy-sz): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 31 March 2021, <www.dmp.wa.gov.au/ens>.

Lu, Y, Wingate, MTD and Morin-Ka, S 2018, 205570: feldspathic sandstone, Sydney Heads Pass; Geochronology Record 1514: Geological Survey of Western Australia, 5p.

Ludwig, KR 2009, Squid 2.50, A User's Manual: Berkeley Geochronology Centre, unpublished report, 95p.

Pirajno, F, Hocking, RM, Reddy, SM and Jones, AJ 2009, A review of the geology and geodynamic evolution of the Palaeoproterozoic Earaheedy Basin, Western Australia: Earth-Science Reviews 94, p. 39–77.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Wingate, MTD, Lu, Y, and Fielding, IOH 2022, Introduction to geochronology information 2022: Geological Survey of Western Australia, 8p.

Recommended reference for this publication

Fielding, IOH, Wingate, MTD, Fletcher, IR, Rasmussen, B, Zi, J-W and Johnson, SP 2022, 195833: sandstone, Sydney Heads Pass; Geochronology Record 1863: Geological Survey of Western Australia, 7p.

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